Local Observation Indices and Validated Period Certificates for Quaternionic Division Points

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M039

October 8, 2026

Abstract

We study the local differential and interpolation interfaces of *Quaternionic division points on curves in the Siegel threefold*. For an ordinary differential system we refine the source's pivot-minor order bound using the largest Smith exponent of its derivative-row lattice and identify an exact finite observation index. We derive two-argument jet formulas, a cancellation-sensitive divided-difference tail bound under an explicitly matched-chart hypothesis, and complete period-pairing error budgets. A fixed graph and ideal-membership certificate tolerates a quantified nonzero analytic residual; its order of required accuracy is tied to the graph denominator and interpolation order. Exact algebraicity and the weighted product formula remain indispensable for the auxiliary polynomial to vanish. The arithmetic common relations, all-branch nonidentity, auxiliary markings, global height bound, and counting-to-finiteness argument are not independently certified here. Contribution assessment: C3/5, because the report proves source-specific local order and precision refinements but no new global quaternionic finiteness theorem.

1 Scope, spaces, and the source claim

The catalogue assigns applicability **L3**. The independent contribution assessment is **C3/5**, for the local refinements proved below. These are distinct scales. "Source" means the public September 24, 2026 manuscript [1]; "local" means a proof in this report; "conditional" identifies a premise not established by our finite computation or scoped reading.

The source claims that a reduced irreducible closed Hodge-generic curve $C \subset \mathcal{A}_{2,\overline{\mathbb{Q}}}$ contains only finitely many points s for which

$$\mathrm{End}^0(A_s) := \mathrm{End}_{\overline{\mathbb{Q}}}(A_s) \otimes \mathbb{Q}$$

is an indefinite quaternion *division* algebra over $\mathbb{Q}$. The condition is equality of the full geometric algebra, not a quaternion embedding into a larger algebra; it excludes $M_2(\mathbb{Q})$. The quaternion algebra, discriminant, and order may vary. No boundary or reduction condition appears in that claim. We neither reprove nor independently certify this global assertion.

On a fixed fine-level family, the source uses the rank-five local system

$$V = \{T \in \mathrm{End}(H^1) : T^\dagger = T,\ \mathrm{Tr}\, T = 0\}, \qquad b(T,U) = \tfrac{1}{4}\,\mathrm{Tr}(TU), \quad q(T) = b(T,T). \tag{1}$$

The adjoint is for the alternating polarization on H^1. In symplectic coordinates,

$$T = \begin{pmatrix} A & \beta J_2 \\ \gamma J_2 & A^{\mathsf{T}} \end{pmatrix}, \quad A = \begin{pmatrix} a & b_0 \\ c & -a \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad T^2 = (a^2 + b_0 c - \beta\gamma) I_4.$$

Thus q has real signature $(3,2)$; it is not a positive error norm. At a division-QM point the actual trace-free Rosati-symmetric endomorphisms span a positive rational two-plane E_t. The lattice $E_t \cap V_{\mathbb{Z},t}$ is saturated, but an arbitrarily selected integral spanning pair need not generate it. Analytic conditioning of a two-column matrix does not prove lattice saturation.

The period line belongs to the three-dimensional quadric $Q_V = \{[z] \in \mathbb{P}(V_{\mathbb{C}}) : q(z) = 0\}$ and is orthogonal to E_t. Horizontal transport $Y(a)$ is directed *back to the center*, with $Y(0) = I$. We always derive a differential equation for the chosen direction; a forward transport matrix and its inverse obey different equations. In local proofs below $y' = A(z)y$ is the declared column convention. Matrix entries or inverse transport can be vectorized to fit it.

1.1 Literal reading and architectural map

The nine assigned TeX files contain 3,282 lines. We read 2,415 distinct lines in bounded untruncated displays: six complete files and three partial files. Two applicable private originals were read over 493 distinct lines, not in full. Their included subfiles were not implicitly read. The 477-line source bibliography was hashed but not literally read. A targeted local corpus search identified candidate paths only; those hits receive no proof-reading credit. Exact intervals, hashes, and unread complements are in the owned read ledger. No whole-source or whole-corpus proof audit is claimed.

Source	Dependency role	Actual boundary
Paper, preamble, figure	Assembly, notation, graph-fiber distinction.	All 118 lines read.
Introduction	Precise QM locus, theorem, height-to-counting strategy.	All 148 lines read.
Geometry	Rank-five quadratic model, fixed covers, full/product monodromy and finite isogeny exceptions.	Lines 84–235 read; remaining proofs imported.
Interpolation	Ordinary-point order bound, constrained graph, successive layers, weighted product formula and descent.	Entire 771-line section read; external effective algebra not reaudited.
Arithmetic	Integral endomorphisms, local comparison, Frobenius grouping, coefficient heights and tails.	Lines 1–165 and 312–775 read; coordinate-height proof 166–311 uninspected.
Height	Auxiliary marking, exceptional isogeny sign obstruction, all-branch nonidentity, height descent.	Lines 1–23, 140–164, 290–396, 545–644 read; key marking/nonidentity proofs imported.
Counting	Flat-coordinate heights, projected path counting, conic incidence and monodromy contradiction, Northcott.	Entire 342-line section read; cited counting and norm theorems imported.

Parent snapshot authentication is provenance, not an author-run proof check or a claim about live branches. Our pre-read and final hashes identify the actual local texts. Parent reading and finite tests are not counted as author work. The only private transfers are the variable-coefficient local moment and derivative rules [6], and the validated ODE, matrix-error, and lattice-gap certificates [5]. These supply no arithmetic comparison theorem.

2 The precise interpolation interface

Let $a_* = x(t) \neq 0$ be an algebraic local parameter value and U_* the auxiliary tuple. Fixed data include curves, ordinary centers, regular frames, connection ranks, and polynomial-combination degrees. "Ordinary" means no pole in the differential system; it does not mean ordinary reduction of the abelian surface. Put $h, d \geq 2$ and $\Psi(h) = 1 + (\log h)^c$, with fixed $c \geq 1$.

For a number field F, local absolute values are the usual *unsquared* ones, with

$$w_v = [F_v : \mathbb{Q}_v]/[F : \mathbb{Q}], \qquad \mathrm{h}(z) = \sum_v w_v \log \max(1, \|z\|_v), \quad \|z\|_v = \max_i |z_i|_v.$$

A complex place has weight $2/[F : \mathbb{Q}]$. Polynomial height uses the complete coefficient tuple with an additional 1, so coefficient scale is controlled. Equivalently, use embeddings with weights $1/[F : \mathbb{Q}]$. On a field extension take *all* extensions of each selected embedding; their total weight is unchanged.

The source theorem assumes polynomially bounded ambient degrees, coefficient heights $Q(d)\Psi(h)$, and $\mathrm{h}(a_*, U_*) \leq Q(d)h$. For a finite weighted selection $\mathcal{V}$ it requires

$$S = \sum_{v \in \mathcal{V}} w_v \log |a_*|_v^{-1} \geq h/Q_1(d), \qquad \sum_{v \in \mathcal{V}} w_v \log^+ \|U_*\|_v \leq Q_2(d)\Psi(h). \tag{2}$$

Here $|a_*|_v < 1$; in the two-argument case $b_* = a_* R_*$ and $|R_*|_v = 1$. A *common*, fixed-length list

$$E_\nu(a, U) = \sum_{j \geq 0} a^j e_{\nu j}(U)$$

vanishes exactly at the selected local evaluations, has $\deg e_{\nu j} \leq C(j+1)$, and has coefficient heights through order l at most $Q_3(l, d)\Psi(h)$. It has convergent tails

$$\left| \sum_{j \geq l} a_*^j e_{\nu j}(U_*) \right|_v \leq |a_*|_v^l e^{(l+1)b_v}, \qquad \sum_{v \in \mathcal{V}} w_v b_v \leq Q_4(d)\Psi(h). \tag{3}$$

Finally the list is not identically zero on *every* normalization branch above every generic component of $a = 0$ in every containing irreducible variety used in descent, whenever a is nonconstant there. Nonidentity at one ordinary point on the original curve is not a substitute for this condition.

Under these hypotheses the source concludes $h \leq C_1 d^{C_2}$. The full interpolation proof was read, including its quotient-connection argument, graph saturation, layer count, and product-formula step. That reading does not verify the effective Gröbner, ideal-membership, and arithmetic Bézout results cited there or the arithmetic supply of its hypotheses.

Why the graph is used. After selecting a combination E nonzero on all relevant branches, take m above its orders divided by the branch orders of a, and put $f = E_{<m}$. The closure of $q = f/a^m$ is formed by saturation away from $a = 0$. The source proves

$$\dim V(I(W'), a, q + e_m) \leq \dim W - 2.$$

This is not an assertion that the graph boundary alone has that dimension: for $q = u/a$ on the (a, u)-plane, its boundary is the whole q-line, and imposing $q + g(u) = 0$ selects a point.

The successive-layer construction produces a polynomial $\mathcal{P}$ of degree at most L, nonzero on W', with a controlled identity

$$\mathcal{P} = \sum_i A_i G_i + A F_n + a^n B_0, \qquad F_n = q + \sum_{j=0}^{n-1} a^j e_{m+j}(U), \quad n = rL. \tag{4}$$

The constrained fiber has dimension at most $k-2$, giving at most $Cn(1+L+C'n)^{k-2}$ conditions against $\binom{L+k}{k}$ independent source polynomials. Fixed ambient dimension and the absence of same-layer substitution are essential. The source fixes the graph first, then r, then L, n, and only then the height threshold. Selected places exploit the membership identity; outside them one evaluates $\mathcal{P}$ itself, whose degree is L, not the potentially larger degrees of A, B_0.

3 Saturated derivative rows and exact observation indices

This is a local refinement of the source's own elementary-divisor argument, not a new general multiplicity principle. Work over $\mathcal{O} = \mathbb{C}[[z]]$, its fraction field $K = \mathbb{C}((z))$, and $\delta = d/dz$. Let $A \in M_N(\mathcal{O})$, $r_0 \in \mathcal{O}^{1\times N}$, and set

$$y' = Ay, \qquad r_{i+1} = r_i' + r_i A, \qquad E = r_0 y. \tag{5}$$

Then $E^{(i)} = r_i y$. Assume $r_0 \neq 0$ and let $\rho \leq N$ be the first index with r_ρ dependent over K on its predecessors. Write $\mathcal{R}$ for the matrix with rows $r_0, \dots, r_{\rho-1}$,

$$M = \operatorname{span}_K(r_0, \dots, r_{\rho-1}), \quad \Lambda_0 = \sum_{i<\rho} \mathcal{O} r_i, \quad \Lambda = M \cap \mathcal{O}^{1\times N}.$$

The quotient Λ/Λ_0 is torsion; write its Smith exponents $0 \leq s_1 \leq \cdots \leq s_\rho$.

Theorem 3.1 (Largest-exponent order bound)**.** *For every formal horizontal solution with $E \not\equiv 0$,*

$$\operatorname{ord}_0 E \leq s_\rho + \rho - 1. \tag{6}$$

Let d_j be the minimum valuation of a nonzero j-minor of $\mathcal{R}$, with $d_0 = 0$. Then $s_\rho = d_\rho - d_{\rho-1}$. In particular this bound is no worse than $\operatorname{ord}_0 \Delta + N - 1$ for any nonzero maximal minor Δ.

Proof. Dependence of the first $\rho+1$ rows makes M stable under $r \mapsto r' + rA$, hence all later rows lie in M. The common kernel in $\mathcal{O}^N$ is saturated and therefore a direct summand over this discrete valuation ring. It is stable under the ordinary connection: for a section v in the kernel, $r(v' - Av) = -(r' + rA)v = 0$, and this derivative is regular. The quotient is free with an ordinary connection. The image $\bar{y}$ of a horizontal solution with $E \neq 0$ is nonzero and has nonzero constant term; otherwise uniqueness of the ordinary coefficient recurrence would make it zero. Its integral dual is Λ. Some $\lambda \in \Lambda$ therefore has λy a unit. Smith form gives $z^{s_\rho}\Lambda \subseteq \Lambda_0$, so

$$z^{s_\rho}\lambda = \sum_{i<\rho} c_i r_i, \qquad c_i \in \mathcal{O}.$$

At least one $r_i y$ has order at most s_ρ. Since $\operatorname{ord} E \leq \operatorname{ord} E^{(i)} + i$ when that derivative is nonzero, (6) follows. Smith reduction by invertible matrices over $\mathcal{O}$ preserves determinantal ideals and gives $d_j = s_1 + \cdots + s_j$. Every maximal minor has valuation at least $d_\rho \geq s_\rho$. □

Theorem 3.2 (Exact finite observation index)**.** *Let Y solve $Y' = AY$, $Y(0) = I$, and define the jet matrix*

$$H_m = \begin{pmatrix} r_0(0) \\ r_1(0) \\ \vdots \\ r_m(0) \end{pmatrix}.$$

There is a least ν with $\operatorname{rank}_{\mathbb{C}} H_\nu = \rho$, and

$$\nu \leq s_\rho + \rho - 1, \qquad \nu = \max_{c:\, r_0 Y c \not\equiv 0} \operatorname{ord}_0(r_0 Y c).$$

For a given exact c, $H_\nu c = 0$ if and only if $r_0 Y c \equiv 0$. All these are exact rank statements, not floating singular-value heuristics.

Proof. The horizontal sections of the saturated kernel have dimension $N - \rho$ and are uniquely specified by their initial values. They are exactly the initial data invisible to all derivatives of E. By Theorem 3.1, any visible solution has a nonzero derivative by index $s_\rho + \rho - 1$, so the kernel of that finite jet matrix is already the invisible initial subspace. Its rank is ρ, proving existence and the bound. If $\nu > 0$, choose an initial vector killed by $H_{\nu-1}$ but not by H_ν; its observable has order exactly ν. The case $\nu = 0$ follows from any nonzero observation at the center. □

A strict improvement in a finite example. Take $A = 0$, $N = 2$, and $r_0 = (z^u, z^v)$ with integers $1 \le u < v$. Then $\rho = 2$, the determinant of the first two rows is $(v-u)z^{u+v-1}$, and their Smith exponents are $(u-1, v)$. The source-style pivot bound is $u + v$, the largest-exponent bound is $v + 1$, and the exact index is $\nu = v$ because $E = c_1 z^u + c_2 z^v$. For $(u, v) = (3, 7)$ these bounds are $10, 8, 7$. This is a manufactured ordinary system, not a new family of QM points.

Gauge and branch qualifications. For a regular invertible gauge $y = G\widetilde{y}$, the new connection is $\widetilde{A} = G^{-1}AG - G^{-1}G'$ and the new rows are $r_i G$. Consequently the Smith exponents and ν are unchanged; the finite jet matrices change by $G(0)$. A singular gauge is excluded. On a branch with $e = \operatorname{ord}_0 a > 0$, an exact observation index gives the sufficient graph choice $m = \lfloor \nu/e \rfloor + 1$ for that expression. A common m must work on every relevant branch after the source's noncancelling combination is chosen. Our local improvement does not provide those branch data, their arithmetic heights, or a uniformly smaller global polynomial $M(D)$.

4 Two-argument jets and local moment coordinates

Write $A(z) = \sum_{k \ge 0} A_k z^k$ and $Y(z) = \sum_{j \ge 0} Y_j z^j$ for the column convention $Y' = AY$, $Y_0 = I$. The exact triangular recurrence is

$$(j+1)Y_{j+1} = \sum_{k=0}^{j} A_k Y_{j-k}. \tag{7}$$

For instance

$$Y_1 = A_0, \quad Y_2 = (A_1 + A_0^2)/2, \quad Y_3 = (2A_2 + 2A_1A_0 + A_0A_1 + A_0^3)/6.$$

The order of noncommuting factors matters. A frozen exponential e^{zA_0} omits the derivative terms. A factorial tail for ordered Volterra iterates is not, in general, a factorial tail in Taylor degree for a variable-coefficient system.

Let $J_0 = I$ and $J_{k+1} = J_k' + J_k A$, so $y^{(k)} = J_k y$. The finite differential observation coordinates $(r_0 y, (r_0 y)', \dots, (r_0 y)^{(m)})$ are therefore an exact finite Ram jet representation with rows r_i; the coefficient derivatives are part of the representation, not optional errors. The ordinary coefficient recurrence and this derivative-row recurrence describe the same jets in normalized and unnormalized coordinates, respectively.

The private local-moment theorem has a compatible, but different, meaning. For an actual output-indexed probe $\mathcal{T}f(x) = \int k_x(u) f(x-u)\, du$, let $m_j(x) = \int u^j k_x(u)\, du$ and $M_j(x) = \int |u|^j |k_x(u)|\, du$. Under its integrability and segmentwise C^{N+1} assumptions,

$$\mathcal{T}f(x) = \sum_{j=0}^{N} \frac{(-1)^j m_j(x)}{j!} f^{(j)}(x) + r_N, \qquad |r_N| \le \frac{M_{N+1}(x)}{(N+1)!} \sup \|f^{(N+1)}\|.$$

Differentiating this expression differentiates the moments too. For source-indexed probes, derivatives instead act on the product of moment and signal. Neither formula produces a period measurement unless that measurement really has the declared integral form. Polynomial exactness does not imply closure or invertibility of a finite coefficient matrix; the private example $I - xD$ kills x.

4.1 The general paired relation

In fixed frames write G, G' for the matrices of the centered trace pairings, and B, P, B', P' for the two-column centered/target pairs. The source relation is

$$E(a, U) = B^{\mathsf{T}} G Y(a) P - B'^{\mathsf{T}} G' Y^{\langle 2 \rangle}(aR) P'. \tag{8}$$

The superscript $\langle 2 \rangle$ means the second connection, not a derivative. The primed data are simultaneously conjugated *labelled* data; they are not generally equal to the unprimed ones. Holding the auxiliary indeterminates fixed during the formal a-expansion gives

$$e_j(U) = B^{\mathsf{T}} G Y_j P - B'^{\mathsf{T}} G' Y_j^{\langle 2 \rangle} P' R^j.$$

This explains the linear growth in the polynomial degree in R. On a normalization branch where $a = a(z)$ and $R = R(z)$ vary, the pullback connection for the second block contains

$$(aR)'(z) A^{\langle 2 \rangle}(aR) = (a'R + aR') A^{\langle 2 \rangle}(aR).$$

Treating R as constant during *that* differentiation is incorrect. Algebraic factors such as $P(z)$ belong in the row and their derivatives are included by (5). Direct sums and fixed tensor constructions put all allowed expressions into the local model of Section 3.

4.2 A cancellation-sensitive matched-chart refinement

For an analytic matrix Y, define

$$D_R(a) = \sum_{j \geq 1} Y_j a^{j-1} S_j(R), \qquad S_j(R) = 1 + R + \cdots + R^{j-1}. \tag{9}$$

Then

$$Y(a) - Y(aR) = a(1 - R) D_R(a).$$

This is a removable divided difference: $D_1(a) = Y'(a)$ and $D_R(0) = Y_1$. It avoids subtracting two nearly equal approximations.

Proposition 4.1 (Matched-pair tail bound)**.** *Suppose Y is holomorphic on a neighborhood of $|z| \leq r$ and $\sup_{|z| \leq r} \|Y(z)\| \leq M$. For $|a| \leq s < r$, $|R| \leq 1$, and $N \geq 1$, truncate (9) at $j = N$, obtaining $D_R^{[N]}$. With $t = s/r$,*

$$\|D_R(a) - D_R^{[N]}(a)\| \leq \frac{M}{r} \frac{t^N((N+1) - Nt)}{(1-t)^2}. \tag{10}$$

Consequently the error in $Y(a) - Y(aR)$ is at most this bound times $|a||1 - R|$.

At a nonarchimedean place, if instead the coefficient bounds $\|Y_j\|_v \leq M_v r_v^{-j}$ are supplied, then for $|a|_v \leq s_v < r_v$, $|R|_v \leq 1$ the corresponding bound is $M_v r_v^{-1}(s_v/r_v)^N$.

Proof. Cauchy's formula gives $\|Y_j\| \leq M r^{-j}$, and $|S_j(R)| \leq j$. Sum the derivative of the geometric tail: $\sum_{j>N} j t^{j-1} = t^N((N+1) - Nt)/(1-t)^2$. At a finite place, $|S_j(R)|_v \leq 1$ by the ultrametric inequality; the supremum of the decreasing coefficient bounds replaces the sum. Neither finite-place assertion is inferred from the complex Cauchy bound. □

For (8), this direct factorization applies only if the two connection charts and the centered pairing rows have been exactly identified. If they are, put $L = B^{\mathsf{T}}G$; the identity

$$LY(a)P - LY(aR)P' = a(1-R)LD_R(a)P + LY(aR)(P-P') \tag{11}$$

separates the cancellation gain from the label mismatch. There is no factor $1-R$ for the second term. Generic conjugate charts in the source need not admit this reduction; it is an explicit conditional computational improvement, not a simplification of all arithmetic groups. Dividing a common relation by $1-R$ without proving its nonvanishing and controlling its local sizes would change the interpolation problem.

5 Validated period and pairing budgets

All complex norms in this section are Euclidean vector or induced spectral norms unless a positive metric is displayed. The trace form G is allowed to be indefinite and is never silently used as a norm.

Proposition 5.1 (Ordinary Taylor and defect bounds)**.** *If A is holomorphic near $|z| \le r$, $\|A(z)\| \le M_A$, and $Y(0) = I$, then*

$$\|Y(a) - \sum_{j=0}^{N} Y_j a^j\| \le e^{M_A r} \frac{(|a|/r)^{N+1}}{1-|a|/r}, \qquad |a| < r.$$

Separately, on a real path parameter $t \in [0,T]$, let $y' = A(t)y$, let p be differentiable, and set $r_p = p' - Ap$. For a differentiable positive Hermitian metric $H(t)$ define

$$\mu_H(t) = \frac{1}{2}\lambda_{\max}\left(H^{-1/2}(H' + A^*H + HA)H^{-1/2}\right).$$

Then

$$\|y(t) - p(t)\|_{H(t)} \le e^{\int_0^t \mu_H} \|y(0) - p(0)\|_{H(0)} + \int_0^t e^{\int_s^t \mu_H} \|r_p(s)\|_{H(s)}\, ds. \tag{12}$$

Proof. Integrating along radial segments and applying Gronwall bounds $\|Y\|$ by $e^{M_A r}$ on the closed disk. Cauchy coefficient bounds and a geometric sum prove the first assertion. For a homogeneous error, differentiate v^*Hv to get growth at most $2\mu_H v^*Hv$; variation of constants gives (12). □

The metric derivative H' and the gauge term $-G^{-1}G'$ cannot be omitted. For a complex path $z = \gamma(t)$ use $A_{\text{path}}(t) = \gamma'(t)A(\gamma(t))$. A chain of panels must carry its local-system basis and connection transitions; two nearby endpoint matrices do not identify monodromy branches.

A useful local improvement in the bound. For a skew-Hermitian path connection and $H = I$, one has $\mu_H = 0$, so the error is initial error plus the integrated defect. The bound using $\|A\|$ instead would multiply by an exponential. This is a standard energy estimate from the private period argument, applied with the source's frame and direction conventions; it is not new arithmetic input. For general frames, outward-rounded enclosures must justify the chosen upper bound for μ_H.

Proposition 5.2 (Complete product perturbation budget)**.** *For matrices $X_1 \cdots X_k$ and approximations $\widehat{X}_i$ of matching sizes, with $\|X_i - \widehat{X}_i\| \le \epsilon_i$,*

$$\|X_1 \cdots X_k - \widehat{X}_1 \cdots \widehat{X}_k\| \le \sum_{i=1}^{k} \left(\prod_{j<i}(\|\widehat{X}_j\| + \epsilon_j)\right) \epsilon_i \left(\prod_{j>i} \|\widehat{X}_j\|\right). \tag{13}$$

Apply this with $(X_1, X_2, X_3, X_4) = (B^{\mathsf{T}}, G, Y(a), P)$. *For a paired relation, add the two product budgets and the certified final subtraction error. For a constant relation, also include the error in its prescribed trace matrix.*

Proof. Telescope using true factors before the replaced factor and implemented factors after it, then apply submultiplicativity. The ordinary transpose has the same spectral norm. □

The Y error includes coefficient, truncation, path, initial-data, and evaluation errors. If a, R themselves are uncertain, an evaluation bound must cover their product: for example $|aR - \widehat{a}\widehat{R}| \le |\widehat{R}|\epsilon_a + |\widehat{a}|\epsilon_R + \epsilon_a\epsilon_R$, followed by a derivative bound on a containing panel. The last mixed term is essential. A displayed number without outward rounding is not such an enclosure.

If a coordinate decoding step solves $C = B_0^{-1}P_0$, assume $\|B_0 - \widehat{B}_0\| \le e_B$, $\|P_0 - \widehat{P}_0\| \le e_P$, and $\|\widehat{P}_0 - \widehat{B}_0\widehat{C}\| \le e_s$. For a certified $s_B \le \sigma_{\min}(\widehat{B}_0)$ with $s_B > e_B$,

$$\|C - \widehat{C}\| \le \frac{e_P + e_B\|\widehat{C}\| + e_s}{s_B - e_B}. \tag{14}$$

Subtract $B_0\widehat{C}$ from P_0 and use singular-value perturbation to prove this bound. Finite rank certificates follow only for the declared finite space. The source deliberately avoids adjoining inverse target Gram matrices to its arithmetic tuple: a numerical inverse may be convenient computationally but carries height and denominator costs absent from its theorem.

6 A robust fixed graph certificate

This section changes a single local interpolation step under explicit hypotheses. It is not a theorem replacing exact common relations throughout the full adaptive descent.

Fix a number field, an algebraic tuple (a, U) with $a \neq 0$, a common expression $E = \sum_{j\ge 0} a^j e_j(U)$, and integers $m, n \ge 1$. Set

$$f = \sum_{j<m} a^j e_j(U), \quad q = f/a^m, \quad F_n = q + \sum_{j=0}^{n-1} a^j e_{m+j}(U).$$

The graph point $z' = (a, U, q)$ is algebraic. No assumption that $E(z')$ is algebraic is made.

Lemma 6.1 (Exact residual identity)**.** *At every selected convergent local evaluation,*

$$F_n = a^{-m}E - a^{-m}\sum_{j\ge m+n} a^j e_j(U). \tag{15}$$

If $\|E\|_v \le \epsilon_v$ *and the tail satisfies* (3)*, then, for a scalar expression,*

$$|F_n|_v \le c_v \max\{|a|_v^{-m}\epsilon_v, |a|_v^n e^{(m+n+1)b_v}\}, \quad c_v = \begin{cases} 2 & v \mid \infty, \\ 1 & v \nmid \infty. \end{cases} \tag{16}$$

In particular, if

$$\epsilon_v \le |a|_v^{m+n} e^{(m+n+1)b_v}, \tag{17}$$

the same order-n gain as for an exact relation holds, up to c_v.

Proof. Multiplying F_n by a^m gives exactly the truncation of E through degree $m + n - 1$. Subtract this from E and apply the ordinary or ultrametric triangle inequality. This proves the identity and both bounds. □

Also

$$|q|_v \le c_v \max\{|a|_v^{-m}\epsilon_v, e^{(m+1)b_v}\}.$$

Thus errors in E affect the graph coordinate as well as the shifted relation. Condition (17) alone does not preserve the original $(m+1)b_v$ coordinate bound; a safe bound under it is $c_v e^{(m+n+1)b_v}$ when $|a|_v < 1$. Alternatively the stronger requirement $\epsilon_v \le |a|_v^{m+n} e^{(m+1)b_v}$ preserves the smaller coordinate budget up to c_v. These costs must enter the evaluation of the membership coefficients.

Theorem 6.2 (One-step algebraic zero certificate)**.** *Suppose the exact graph equations $G_i(z') = 0$ and exact identity (4) are certified. Let $\mathcal{P}$ have algebraic coefficients and let $\mathcal{W}$ be a finite weighted selection of places or embeddings. For each $v \in \mathcal{W}$ assume $|a|_v < 1$, (17), and certified bounds*

$$|A(z')|_v \le e^{u_v}, \quad |B_0(z')|_v \le e^{v_v}, \qquad u_v, v_v \ge 0.$$

Assume also a valid outside-place bound for a nonzero value,

$$\sum_{v \notin \mathcal{W}} w_v \log |\mathcal{P}(z')|_v \le H_{\text{out}}.$$

Then $\mathcal{P}(z') = 0$ if

$$n \sum_{v \in \mathcal{W}} w_v \log |a|_v^{-1} > H_{\text{out}} + \sum_{v \in \mathcal{W}} w_v \left[2 \log c_v + \max\{u_v + (m+n+1)b_v, v_v\}\right]. \tag{18}$$

Proof. At the target, the membership identity is $\mathcal{P}(z') = A(z')F_n(z') + a^n B_0(z')$. Lemma 6.1 and one further triangle inequality give

$$\log |\mathcal{P}(z')|_v \le n \log |a|_v + 2 \log c_v + \max\{u_v + (m+n+1)b_v, v_v\}.$$

If the algebraic value were nonzero, sum this inequality and the outside bound with the normalized weights. The product formula makes the resulting left side zero, while (18) makes the right side negative. □

One universally available conservative outside bound is $H_{\text{out}} = \mathrm{h}(\mathcal{P}) + L\mathrm{h}(z') + \log \#\,\mathrm{mon}(\mathcal{P})$ for $\deg \mathcal{P} \le L$, using the complete affine coefficient height. A sharper bound may retain only outside-place contributions. In either case the algebraic coefficients, exact ideal membership, graph coordinate, and their heights are prerequisites; an approximate identity of polynomials does not satisfy the theorem.

Selection loss and feasible precision. If only a subset $\mathcal{W} \subseteq \mathcal{V}$ meets the precision threshold, set $S_{\text{lost}} = \sum_{v \in \mathcal{V} \setminus \mathcal{W}} w_v \log |a|_v^{-1}$. The left side of (18) is $n(S - S_{\text{lost}})$; excluded places must be charged in H_{out}. All extensions of each retained embedding must still be taken. This is a controlled weakening of a *fixed certificate's* local exact-zero premise, not a free weakening of the global theorem. In particular a fixed decimal residual tolerance is not compatible with arbitrary small a and growing $m+n$. For full descent one would need uniformly validated residuals, graph-coordinate bounds, all-branch nonidentity, coefficient heights, and parameter choices on every new variety. None is supplied by the local theorem.

7 Exactness gates and the global arithmetic boundary

7.1 What a small period residual can certify

The source's good-reduction comparison, when its geometric hypotheses are established, identifies

$$b(B_i, Y(a)P_j) = \tfrac{1}{4}\,\mathrm{Tr}(\beta_i \alpha_j) \in \tfrac{1}{4}\mathbb{Z}.$$

The endomorphisms are actual integral endomorphisms of the common special fiber. At infinity, the analogous statement uses integral Betti transport. The scalar quarter-lattice is a conclusion of these exact geometric identifications, not a property of arbitrary de Rham coordinate vectors.

Proposition 7.1 (Conditional archimedean trace isolation)**.** *Suppose exact geometric data establish that a real trace residual ξ lies in $\frac{1}{4}\mathbb{Z}$. If a validated computation produces $\widehat{\xi}$ with $|\xi - \widehat{\xi}| \le \epsilon$ and $|\widehat{\xi}| + \epsilon < 1/4$, then $\xi = 0$. The same assertion applies entrywise to a matrix residual when the full entrywise budget is strictly below $1/4$.*

Proof. Every nonzero element of $\frac{1}{4}\mathbb{Z}$ has ordinary absolute value at least $1/4$, contradicting the certified strict inequality. □

This can identify a prescribed trace relation after its arithmetic membership premise is proved. It does not establish the quaternionic endomorphism algebra, identify a new integral endomorphism, or show equality of two conjugate systems at other places. There is no corresponding positive gap for the p-adic norm on $\frac{1}{4}\mathbb{Z}$: the nonzero elements $p^k/4$ tend to zero p-adically. A nearest-quarter reconstruction also needs an error less than $1/8$ for unique identification of an unspecified quarter-integer.

The private finite-space certificate generalizes this distinction. If $v \in D^{-1}\mathbb{Z}^b$ is *already known* to have that common denominator, and a test map on the whole comparison space obeys $\sigma_{\min}(T) \ge s > 0$, then

$$\|Tv\| < s/D \quad \Longrightarrow \quad v = 0.$$

For uncertain T, the residual bound includes operator error times a bound for $\|v\|$. Without the denominator hypothesis, the vectors e_1/n refute any fixed positive gap. Without injectivity, a nonzero kernel vector has zero observed residual. An interval containing zero proves neither exact vanishing nor full-space membership. An exact jet-rank certificate in Theorem 3.2 likewise says only what its specified ordinary local system implies.

7.2 How the source supplies its common relations

The arithmetic section distinguishes global frame denominators from restricted local sizes on the center disks. It proves potential good reduction using the impossibility of a nonzero torus character module of rational dimension at most two over a four-dimensional quaternion division algebra. Small integral endomorphisms and their polynomial Rosati bounds require separate endomorphism estimates. The coordinate-height proof was not read in full here and remains imported.

At a good finite place the source's functorial crystalline/de Rham comparison places centered and target endomorphisms in the *same actual* special fiber, allowing ramified coefficient fields. It is not an arbitrary isogeny identification. A Frobenius lift fixes the rational cross-trace matrix; subtracting the conjugate equation gives (8), with simultaneous conjugation of all labels and $|R|_v = 1$. Polynomially many groups of conjugate labelled tuples retain a weighted smallness sum. The normal closure indexes embeddings only; its potentially large degree is not charged to the tuple field or the normalized weights.

At unexceptional primes, the ordinary integral connection gives $Y_j \in (j!)^{-1} M_5(\mathcal{O}_v)$. The source uses the stronger-than- convergence cutoff

$$-\log|a|_v > 2\alpha_p, \qquad \alpha_p = \log p/(p-1),$$

and obtains coefficient bounds $\|Y_j\|_v \le e^{j\alpha_p}$. Its tail is a supremum of decreasing terms, with no constant greater than one charged at every prime. The loss from excluding insufficiently small places is controlled through $\sum_{p:\,|a|_v<1 \text{ for some } v|p} \log p \le [\mathbb{Q}(a):\mathbb{Q}]\mathrm{h}(a)$ and a logarithmic-square estimate for $\sum \alpha_p$. Our complex error norms alone do not prove any of these all-place facts.

7.3 Marking, nonidentity, and counting are separate inputs

The source retains all four entries of the trace-pair equations. On exceptional isogeny correspondences those entries force a common sign on both endomorphism planes. A fixed auxiliary prime and marking provide a saturated rank-four sum of lattices with degenerate reduction. The positive-sign case conflicts with the polarized multiplier p/m; the negative-sign case conflicts with the required orthogonal eigensummand in a unimodular lattice. This requires exact markings, saturation, actual isogenies and their common specialization. We read the statements and setup, not the complete marking or nonidentity proofs. Numerical positive definiteness of a two-plane does not establish those facts.

After the height bound, counting uses a different height: for rational coordinates $y_i = a_i/b_i$ in lowest terms,

$$H_{\mathrm{rat}}(y) = \max_i \max\{|a_i|, b_i\}.$$

This is multiplicative, not the absolute logarithmic height above. Endomorphism and dual Hodge-norm estimates give polynomially bounded *numerical sizes* of ten integral flat coordinates at all conjugates. The source's quoted projected path-counting theorem counts distinct curve arguments and does not presume finite fibers over the rational-vector coordinates.

The final geometric deduction is transparent under those inputs. A one-real-parameter semialgebraic family of nondegenerate pairs has complex Zariski closure of dimension at most one. For each pair, the orthogonal nondegenerate three-space cuts Q_V in a conic. The incidence over the nondegenerate locus therefore has dimension at most two. One closes its image *after* restricting to that locus; arbitrary fibers over degenerate pairs must not be added. A nonconstant period arc in this image forces an analytic identity, whereas full $\mathrm{SO}(3,2)$ monodromy makes its complexified period orbit Zariski dense in the three-dimensional quadric. Bounded degrees then follow, and Northcott applies with bounded ample height. This chain uses the arithmetic height theorem, definability, counting, and monodromy; no local residual certificate replaces them.

8 Practical use, limitations, and future work

For a specified ordinary branch, the proposed finite workflow is: construct the exact derivative rows, determine their generic rank, compute determinantal valuations or Smith data, and locate the first full-rank jet matrix. Use this to avoid unnecessary local Taylor orders when choosing a graph truncation. Separately carry validated complex or finite-place coefficient bounds for the actual charts. Evaluate paired relations using matched-chart cancellation only where the exact identifications hold, otherwise use the full two-block formula and error budget.

For a fixed algebraic graph certificate, combine the actual residual accuracy, graph-coordinate sizes, membership coefficients, outside height, and selected weighted smallness in Theorem 6.2. A positive margin proves an algebraic auxiliary value zero, not the original analytic expression zero. If the margin fails, the result is inconclusive; one must improve the precision, choose another certified certificate, or retain more weighted places. No rounding of a small number supplies a missing algebraicity or integrality premise.

Advantages. The largest Smith exponent is invariant under regular gauge changes and can be strictly smaller than a pivot valuation. The exact observation index removes remaining slack for a fixed ordinary branch. Divided differences reveal a genuine $|a||1-R|$ cancellation where the charts match. The residual identity specifies the otherwise hidden $(m+n)$ precision cost and accounts for growth in the graph coordinate. All are finite, source-specific deductions with explicit proofs.

Costs. Exact Smith or jet computations may require algebraic branch representations unavailable from a numerical period oracle. Singular gauges, poles, and unknown branch changes invalidate the ordinary analysis. Matched charts and matched labels are extra assumptions. Denominator, global height, and common-local- relation premises are arithmetic and can dominate the effort. Even good local conditioning says nothing about effectiveness of the global finiteness bound. Classical zero estimates, Taylor theory, and lattice isolation are not new Ram theorems.

Feasible next steps. A useful bounded project is to implement the exact observation index for a concrete algebraic connection and normalization branch, comparing the three bounds in Section 3. Another is to test the matched-chart tail bound on an explicitly identified isogeny chart with certified transition matrices. A more demanding extension would propagate the residual budget uniformly through every graph in a controlled descent; this requires a new uniformity argument and cannot be inferred from our one-step theorem. Finally, simultaneous archimedean and finite-place certificates should record their normalizations and field-extension weights in one ledger without interpreting a positive archimedean metric as the indefinite trace form.

9 Literature and evidence assessment

An October 8, 2026 primary-source refresh used public URLs only. Bertrand–Beukers [2], Theorem 1 on printed pages 181–182, is an established differential-rank multiplicity estimate for polynomial expressions in solutions of a fixed rational linear system. We read its primary PDF text at the statement, not its complete proof; screenshot retrieval timed out. Our local Smith-lattice proof is supplied independently and is attributed as a refinement of the source's own ordinary-point argument.

Papas [3], whose primary arXiv page records October 22, 2025, studies archimedean and p-adic relations for quaternionic multiplication in one-parameter families of abelian surfaces. Daw–Orr–Papas [4], v2 dated February 25, 2026, concerns curves in $Y(1)^3$, including boundary and limited-supersingular-proximity cases. It must not be cited as a proof of the present unrestricted $\mathcal{A}_2$ statement. For these two references we inspected metadata and abstracts only; no theorem from their unread proofs is an independent premise of our local results. The source's larger cited arithmetic literature remains imported.

The private originals were searched and read locally. No private title, passage, or generated report text was submitted to public search. The actual read intervals are 241–358, 470–553 and 587–626 in the period original, and 1–162 and 194–282 in the local-moment chapter. Adjacent input directives do not extend that scope. The chapter's source-centered and output-centered distinction, derivative product rule, and warning about finite closure were retained, as were the period original's exactness and conditioning restrictions.

Author-owned exact tests cover the rank-five trace normalization, noncommutative Taylor recurrence, Smith/observation example, regular-gauge row identity, divided differences, product-error budgets, graph residual identity, and strict arithmetic-gap boundaries. They are manufactured finite cases, not tests of the source's global arithmetic theorem. The build and all-page visual records, literal-read ledger, proof boundary, identity record, standard assessment list, and freeze are separate owned artifacts. Parent tests are neither recounted nor frozen.

On the five-point contribution scale, C1 denotes no demonstrated increment, C2 limited, C3 notable local, C4 substantial, and C5 very high. We assign **C3**: the observation-index and precision refinements are proved and tied to the source's actual interpolation mechanism. The rank-five model, classical multiplicity principle, generic Taylor error bound, and qualitative arithmetic conclusion are not claimed as new. C4 or C5 would require new uniform arithmetic, global geometric, or effective counting information

that this report does not supply.

References

[1] OpenAI, *Quaternionic division points on curves in the Siegel threefold*, September 24, 2026. Pinned public source snapshot. The owned read ledger gives full local SHA-256 values; source section numbers 00–05 refer to its build files.

[2] D. Bertrand and F. Beukers, *Équations différentielles linéaires et majorations de multiplicités*, Ann. Sci. École Norm. Sup. (4) 18 (1985), 181–192, Theorem 1. Primary Numdam PDF.

[3] G. Papas, *On the v-adic values of G-functions III*, arXiv:2510.19683v1, October 22, 2025.

[4] C. Daw, M. Orr, and G. Papas, *Some new cases of Zilber–Pink in* $Y(1)^3$, arXiv:2510.09603v2, February 25, 2026.

[5] M. Subbarao, private original `res-math/TR-item-011-astra.tex`, selected proof passages as specified in the read ledger. Finite period certificates, variable-coefficient continuation, perturbation and lattice gaps.

[6] M. Subbarao, private original `volume1-rev2/chapters/ch03-local-moments-rev2.tex`, selected proof passages as specified in the read ledger. Local moments, coefficient derivatives, and source localization.